



- PANDAS
- SEABORN
- PLOTLY
- PRATICS
- DATA INSIGHTS

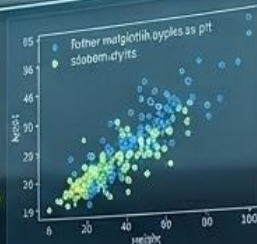
STARTING SOON

```
import matplotlib.pyplot as plt  
  
# Importing data  
data = pd.read_csv('data.csv')  
plt.scatter(data['x'], data['y'])  
plt.show()
```

```
# Data  
plt = matplotlib.pyplot  
plt = matplotlib.pyplot  
plt.show()
```

```
# Seaborn  
plt = sns.relplot(  
    data=data,  
    x='x',  
    y='y',  
    hue='category',  
    style='category',  
    facet_kws=dict(sharex=False)  
)  
plt.show()
```

DATA VISUALIZATION



Data Visualization Camp using Python

22 – 25 June 2026





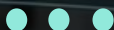
10:00 – 10:15 AM • 15 mins

Choosing your Visual Story

Class begins shortly — please take a seat

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STARTING SOON



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"A chart is worth 1,000 spreadsheet rows"

— Every Data Scientist Ever



Raw Data

Numbers in a table make it hard to see patterns

Most of us cannot process thousands of data points!



Visualization

A chart or graph transforms numbers into shapes you CAN see.

Multiple charts and graphs (and summary numbers) can reveal hidden angles.



Insight

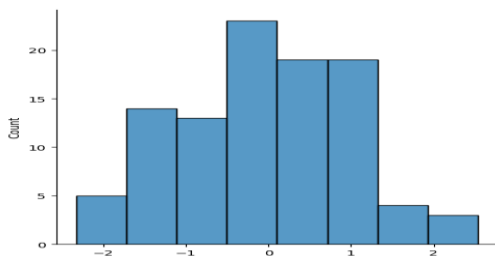
Now you can:

- Tell a story
- Motivate analysis
- Make a decision
- Motivate more questions!

Univariate — One Variable at a Time

"Uni" = ONE. You are looking at how ONE thing is spread out or distributed.

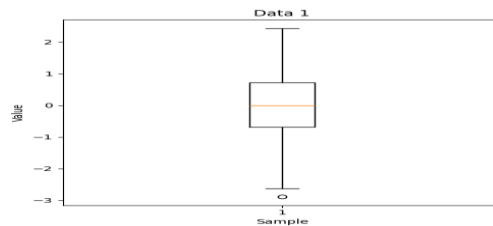
Histogram



Shows how often values appear in ranges

"How are measurement errors distributed?"

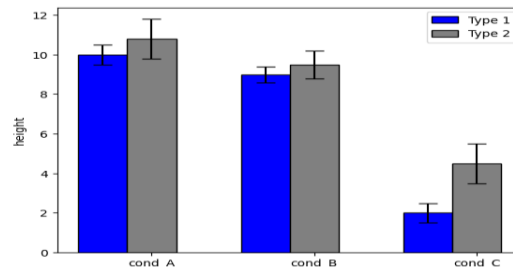
Box Plot



Shows median, spread, and outliers

"How are measurement errors distributed?"

Bar Chart

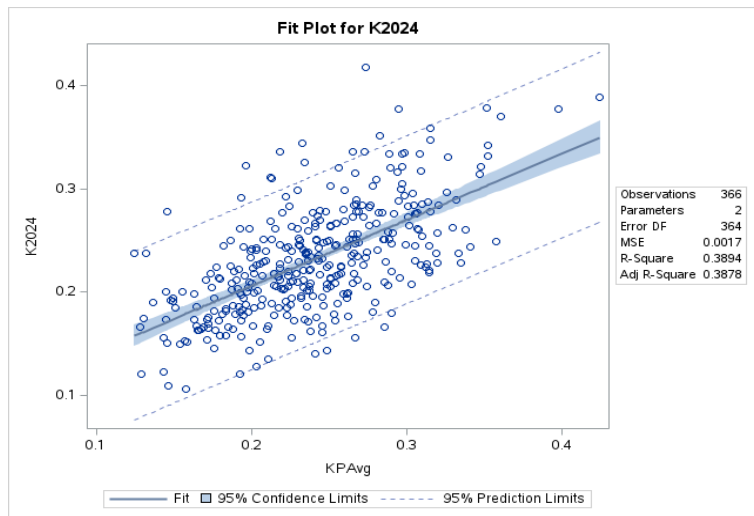


Compare counts across categories

"How do the heights of two crops differ across varying treatment conditions?"

"Bi" = TWO. You are looking at the relationship between two things. Does one affect the other?

Scatter Plot – Relationship

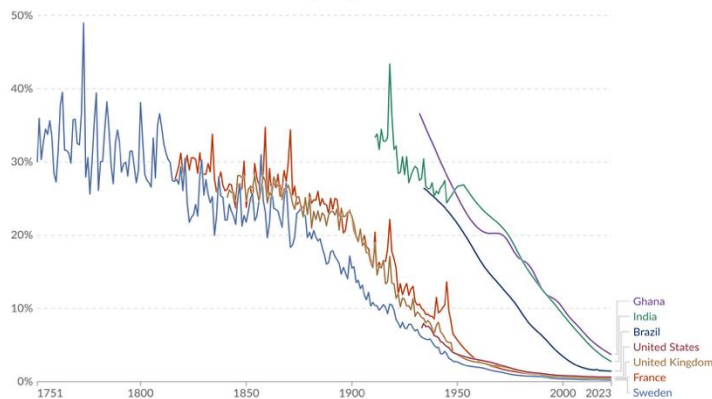


Example: How do predicted strikeouts match reality?

Line Chart – Trend Over Time

Child mortality rate

Estimated share of newborns who die before reaching the age of five.



Data source: Gapminder (2015); UN Inter-agency Group for Child Mortality Estimation (2025) OurWorldinData.org/child-mortality | CC BY

Example: Child mortality over time

Multivariate — Many Variables

"Multi" = MANY. See 3, 4, or 5 variables in a SINGLE chart — advanced & powerful!



Bubble Chart

X, Y axes + bubble SIZE as a 3rd variable

"Country GDP vs Life Expectancy vs Population"



Heatmap

Color intensity shows a 3rd value in a grid

"Which hour of which day is Twitter most active?"



Pair Plot

Every variable vs every other — all at once!

"Which features of Iris flowers relate to each other?"

The Plot Picker

1

If: I want to see how data is spread out

→ Histogram or Box Plot

1-2

If: I want to compare groups/categories

→ Bar Chart or side-by-side bar chart

→ Side-by-side Histogram or Boxplot

2

If: I want to see if two things are related

→ Scatter Plot

2

If: I want to see change over time

→ Line Chart

3+

If: I want to see 3+ variables together

→ Heatmap or Bubble Chart

Many

If: I want to explore all variables at once

→ Pair Plot (Seaborn)

• PRACTICES

Charts You've Already Seen!



NBA Stats App

Scatter plot: Points Scored vs Minutes Played

Bivariate



Weather App

Line chart: Temperature over the past 7 days

Univariate



Spotify Wrapped

Bar chart: Your top genres of the year

Univariate



Stock Market

Line/candlestick: Price change over time

Bivariate



Election Results

Heatmap/choropleth: Results by state

Multivariate



Game Leaderboards

Bar chart: Top players ranked by score

Univariate

Quick Reflection

Day 1

Think about something you care about — sports, music, games, fashion, food...

1

What data would you collect about it?

→ Scores? Plays? Followers?

2

How many variables do you want to compare?

→ One → Histogram, Two → Scatter, Many → Pair Plot

3

What story do you want to tell?

→ Show a trend? Compare groups? Find relationships?

• SEABORN

4

Who is your audience?

→ Coach? Principal? Friends? The internet?



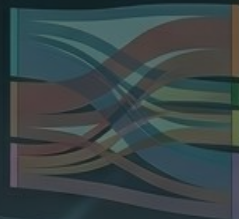
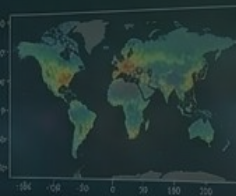
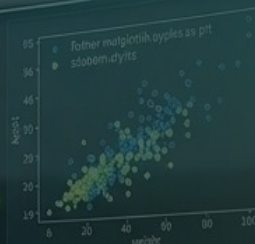
Coming Up: Snack Break! Then — Invited Lecture: The Role of Data in Science

Questions?

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# Import data from a CSV file  
data = pd.read_csv('data.csv')  
  
# Create a scatter plot  
plt.scatter(data['x'], data['y'])  
plt.show()
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